

FLCR-05 - Typed Boundary Repair

Series: Formalizing LCR, Unifying Standard Models **Artifact:** formal paper source **Status:** first-pass rich rewrite; requires final citation and build pass. **Claim posture:** maximal local claim posture with explicit lane boundaries.

Abstract

This paper turns failed joins into typed, replayable repair rows. Boundary repair is not a proof of the desired downstream claim; it is a typed exception and proof-obligation discipline that preserves state, coordinate, cause, route, source, target, and falsifier. The main result is an idempotent normalization of failure into legal next-step data.

Keywords

boundary repair; typed exception; proof obligation; idempotence; schema.

External Reader Orientation

An outside reviewer should read this paper as a formal-system construction paper. Its local object is **Typed Boundary Repair**. The paper's immediate contribution is: Turns failed joins and boundary mismatches into typed repair data.

The nearest external anchors are finite-state reasoning, typed interfaces, proof-carrying records, conservative extension, and ordinary mathematical citation discipline. LCR terms are therefore internal labels for the construction unless a row explicitly imports an external theorem, citation, dataset, or calibration target.

It does not ask the reader to accept any Standard Model or literal-physics identification at this layer. Such mappings are deferred to the translation papers and must carry their own evidence.

Reviewer Compression

Core object. Typed Boundary Repair.

Primary result. This paper contributes the local formal object and its safe downstream-use rule.

Primary non-result. This paper does not claim direct identity with Standard Model or physical objects.

Review strategy. Evaluate the formal claim rows first, then inspect the receipt/citation/calibration rows that support each claim. Appendices provide routing and implementation context; they should not be read as stronger than their lane permits.

Peer Reviewer Assessment

Reviewer assumption: the reviewer has been briefed on the FLCR structure, claim lanes, CAM/crystal routing, forge/action surfaces, and the distinction between internal formal results and external physical calibration.

Recommendation. major revision, technically promising.

Review score vector. orientation=2, claim_granularity=2, evidence_visibility=2, falsifier_visibility=2, journal_apparatus=1, base_layer_boundary=2.

Formal claim rows reviewed. 3.

Reviewer Strength Finding

The manuscript is now readable as a bounded formal-system paper: it identifies its local object, separates internal construction from physical interpretation, and exposes claim lanes for review.

Major Revision Requests

- Convert the strongest proof sketch into numbered definitions, lemmas, and propositions with explicit dependencies.
- Replace placeholder source classes with final citation keys, receipt hashes, or dataset identifiers wherever possible.
- Move repeated pass-derived material into a compact evidence appendix and keep the main body focused on the paper-local argument.

- Add at least one worked example showing the local object, legal operation, receipt, residue, and downstream import rule.

Minor Revision Requests

- Normalize theorem capitalization and punctuation.
- Add final figure/table captions where the text currently describes diagrams schematically.
- Ensure every acronym and local term appears in the paper-local glossary.

Acceptance Blockers

- A reviewer can accept the formal contribution without accepting later physics claims, but only if final citations and receipt identifiers are attached.
- The proof sketch must be promoted into a formal proof or explicitly labeled as a construction argument.

1. Contribution And Scope

- Defines repair rows as typed constraints rather than proof promotions.
- Installs rejection of untyped failures as a formal schema rule.
- Connects repair rows to graph edges for later causal ledgers.
- Routes physical curvature, dust, oloid, and material readings downstream.

This paper belongs to the LCR-native construction band. Physics and Standard Model language, when it appears, is only a deferred mapping cue, analogy, or explicitly bounded calibration target. The paper's base object must remain stable enough for later papers to cite without importing a stronger physical claim by implication.

2. Source Routing

Primary routed sources:

- 04_Boundary_Repair.md
- supplements/04_INTERNAL_REASONING_SUPPLEMENT.md

Cross-corpus evidence classes:

- CAM_CLAIM_100_SCORE_AUDIT.md
- NSIT_TOOL_CLOSURE_MATRIX.md
- NSIT_REASONED_CLOSURE_PASS.md
- PAPER_REWORKS_CRYSTAL_PROJECTION.json
- standards, glue, window, and node databases where applicable
- notebook-derived review prompts and orbital source manifests

3. Definitions

- **Repair row.** A structured obligation record with state, coordinate, reason, route, source, target, boundary, and receipt fields.
- **Typed exception.** A failure promoted into a catchable, inspectable, replayable record.
- **Idempotent repair pass.** A normalizer that leaves already-typed repair rows unchanged.
- **Untyped failure.** A failure marker lacking the fields needed for routing or replay.

4. Formal Claims

Claim	Lane	Statement
Theorem 5.1	receipt_bound_internal_result	A valid boundary repair converts a failed join into a typed proof-obligation row while preserving the relevant coordinates.

Claim	Lane	Statement
Proposition 5.2	receipt_bound_internal_result	Repair is idempotent on already-normalized rows.
Protocol 5.3	normal_form_result	Untyped failure cannot be consumed by later papers as repair evidence.

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 5.1	finite/executable result	A valid boundary repair converts a failed join into a typed proof-obligation row while preserving the relevant coordinates	receipt, validator, executable pass, generated artifact, or finite enumeration	the stated finite/executable domain only	verify the receipt exists and its scope matches the claim
Proposition 5.2	finite/executable result	Repair is idempotent on already-normalized rows	receipt, validator, executable pass, generated artifact, or finite enumeration	the stated finite/executable domain only	verify the receipt exists and its scope matches the claim
Protocol 5.3	formal construction	Untyped failure cannot be consumed by later papers as repair evidence	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR-05 defines or uses Typed Boundary Repair as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	Imported mathematics remains an external theorem/citation target; this paper may use it only through declared mappings and conservative-extension discipline.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Physical or Standard Model identity is not claimed here unless the paper contains an explicit calibration row; the default status is deferred mapping.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

5. Paper-Specific Development

1. Take an untyped failure and reject it as non-consumable evidence.
2. Normalize the failure into a repair row with state, coordinate, reason, route, source, target, and boundary.
3. Run the normalizer again and confirm that an already typed row is unchanged.
4. Export the row as a proof obligation to the causal ledger, not as the proof of the desired claim.

Proof Sketch

The repair operation is a normalization function on failure records. Records lacking required fields are outside the valid repair language. Records containing the required fields are stable under the operation, so the pass is idempotent. The normalized row is useful precisely because it is replayable and bounded, not because it proves the target.

6. Construction

The construction is intentionally staged. First, the paper names the finite or typed object that can be inspected directly. Second, it declares the admissible operations over that object. Third, it records the receipt or source class that allows another paper to consume the result. Fourth, it names the residue that must not be silently promoted.

For this paper, the operative object is `Typed Boundary Repair`. Its safe consumption rule is:

```
source object -> declared operation -> receipt/lane -> downstream import
```

The unsafe consumption rule is:

```
source phrase -> downstream identity claim
```

That second pattern is explicitly rejected by `FLCR-01` and must be rewritten as a claim-lane promotion with evidence.

7. Evidence And Receipts

Evidence source	What it contributes
Paper 04 legacy body	Typed boundary repair theorem, verifier, falsifier, repair contract variants.
Internal supplement 04	Typed exception, proof obligation, Hoare-style pre/postcondition framing.
CAM Claim 100 Score Audit	Paper 04 scored 93 internally closed.
NSIT standards lanes	Claim envelope and content-addressed receipt standards support repair-row normalization.

Appendix D. Resolved-State Closure Pass

This pass removes false restrictions on the paper's claim posture. A row is no longer called open merely because a stronger future claim exists. The satisfied lane is closed now; only the stronger claim that lacks its required adapter, comparator, calibration, or falsifier remains as residue.

Closed Now

Row	Lane	Resolved state	Remaining boundary
paper lift enumeration	<code>receipt_bound_internal_result</code>	8/8 LCR states succeeded; error walls=0; avg TarPit mass=0.514103.	This closes the paper-local finite lift readout, not every stronger physical interpretation.
boundary repair	<code>normal_form_result</code>	The typed boundary repair surface is closed as an admission table for legal next-state repair.	The family closure is a structural lane; measured physics still requires destination-specific calibration.
decade-1 tower	<code>receipt_bound_internal_result</code>	The decade tower is resolved with avg TarPit mass=0.510370 and mass sum=40.829567.	The decade total is an internal tower metric, not a standalone universal physical constant.
family-05 cross-lift comparison	<code>cam_crystal_reapplication_result</code>	Family tower binds <code>FLCR-05</code> , <code>FLCR-15</code> , <code>FLCR-25</code> , <code>FLCR-35</code> ; avg TarPit mass=0.514757; error walls=0.	Cross-lift agreement strengthens routing and comparison; it does not erase paper-local boundaries.
typed repair admission	<code>normal_form_result</code>	Boundary repair is closed as a typed admission protocol.	Specific physical repair events require process-specific data.

Claim Posture After This Pass

`FLCR-05` should now be read as a resolved-state contribution for `Typed Boundary Repair` at its declared lane. The paper may state the strongest claim supported by these rows directly. It should reserve caution only for a specifically named stronger claim whose required evidence is absent.

8. Dependencies And Downstream Use

This paper may be imported by later FLCR papers only through the claim lanes above. The default downstream consumers are:

- FLCR- 11 through FLCR- 18 for window, receipt, admission, CA, algebraic, curvature, residue, and exceptional machinery.
- FLCR- 28 through FLCR- 30 for CAM/crystal, corpus ribbon, and universal normal-form intake.
- FLCR- 31 through FLCR- 40 only after the LCR-native result has been cited and the translation claim has its own evidence lane.

9. Limitations And Falsifiers

- A repair row is an input to future verification, not a proof of the repaired claim.
- Domain-specific repair rows may extend the schema but cannot drop the replay fields.
- Curvature or material interpretations require external equations, data, or calibration receipts.

A falsifier for this paper is any example that satisfies the declared input conditions while violating the stated construction, receipt, or lane boundary. An interpretation that merely wants a stronger downstream conclusion is not a falsifier; it is an obligation routed to a later paper.

10. Publication Readiness Tasks

- Validator identities and receipt hashes are recorded in the manifest or receipt appendix.
- External theorems require final citation entries before journal submission.
- Every theorem, proposition, and protocol must retain a claim lane and evidence boundary.
- The Markdown, LaTeX, and PDF forms are generated from the same canonical source.

Dual Positioning: Story And Formal Carrier

This paper must be read in two synchronized ways. Sequentially, it explains why the next state exists in the corpus story. Formally, it binds that state to accepted mathematics, IRL formalism, code receipts, validators, CAM/crystal lookups, and claim-lane boundaries.

Assigned original ribbon role(s): 04/boundary_action.

Original slot	Family	Lift	Role	Proof form
04	boundary_action	0	order-1 slot-4: type the boundary, repair row, or curvature-demand condition	typed boundary row and next-state admissibility

The formal obligation is to state the strongest valid claim available for this paper without letting interpretation silently change the addressed object. Any author interpretation belongs beside the formalism as a declared relabeling, bridge, or analog, and must be accompanied by the evidence lane that makes the claim consumable by later papers.

State-Bound Proof Contract

This paper receives: state emitted by prior slot 03 and reopened at original slot 04 (Boundary Repair).

It must produce: formal result of original slot 04 plus the same-family lift path toward slot 14.

Semantic transition: type the boundary condition that decides whether the state repairs, reflects, curves, or routes onward.

Accepted formalism targets to bind in the journal rewrite:

- boundary value problems
- typed interfaces and admissibility conditions

- topological boundary/interior distinction
- falsifier rows for failed promotions

Slot	Family	Lift	Produced proof form
04	boundary_action	0	typed boundary row and next-state admissibility

The formal carrier uses these accepted tools while preserving interpretive claims as explicit relabelings, correspondences, or bridges. Claim promotion is admissible only when a receipt, standard citation, CAM/crystal reapplication, normal-form result, calibration datum, or falsifier boundary is attached.

Provenance And Crystal Evidence Integration

This section integrates prior work, CAM/crystal projections, NSIT tooling, standards-window evidence, and routed supplements as provenance-bearing evidence. Source material is not treated as manuscript text; it is bound into the paper only through claim lanes, receipts, validators, normal forms, calibration rows, or explicit falsifier boundaries.

Expansion Rule For This Slot

Use past work to type the boundary: admitted, repaired, reflected, calibration-required, falsified, or routed onward.

Primary Evidence Inputs

Source	Placement reason
04_Boundary_Repair.md	primary assigned source for the paper's formal spine
supplements/04_INTERNAL_REASONING_SUPPLEMENT.md	paper-local reasoning supplement contributing to definitions, proof sketch, and limitations

Secondary And Orbital Evidence Inputs

Source	Placement reason
supplements/RECEIPT_VERIFIER_CATALOG.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
supplements/INTERNAL_REASONING_PYRAMID_INDEX.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
supplements/INTERNAL_REASONING_5P_WINDOW_SUPPLEMENT.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
supplements/INTERNAL_REASONING_7P_WINDOW_SUPPLEMENT.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
supplements/INTERNAL_REASONING_9P_WINDOW_SUPPLEMENT.md	cross-cutting supplement; paper-relevant rows, receipts, and guard language are bound through evidence lanes
CAM_CLAIM_100_SCORE_AUDIT.md	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
NSIT_TOOL_CLOSURE_MATRIX.md	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
NSIT_REASONED_CLOSURE_PASS.md	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
ORBITAL_DATA_CROSS_POPULATION_AUDIT.md	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
AGENT_CRYSTAL_WORK_HARVEST.md	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
PAPER_REWORKS_CRYSTAL_PROJECTION.json	series-wide audit/score/closure evidence; import paper-specific rows and leave global context outside the body
NEW_PAPER_NEEDS.md	route relevant obligations into this paper's burden table: NP-05 Receipt Ecology And 32-Paper Causal Graph
Additional source routes	Complete routing is maintained in the source-placement appendix

Crystal And Standards Evidence To Bind

- Paper Reworks crystal projection: 33 paper markdown files, 9 supplements, 5 queues, 6 lattice-forge unification artifacts, 140 faces, 140 vignettes, and 280 CAM nodes.
- Standards-window intake: 192/192 corpus conformance verdicts PASS; 140/142 obligations have candidate routes; 2/4/8/16/32 window lattice is available for cross-paper reads.
- Agent/crystal harvest: agent-generated memory, runtime standards starter, current runtime build, repo-harvest CAM, notebook-derived bundles, and downloaded crystal payloads are orbital evidence surfaces.
- NSIT inventory baseline: 220 validators, 1786 receipt/data artifacts, 1137 formal-paper-like artifacts, 114 supplements, and 86 memory/CAM artifacts.

Paper-Specific CAM/Score Rows

04	93	internally closed	typed repair constraint, falsifier, Paper 05 payload intake	shared ledger population and domain examples
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Paper-Specific NSIT Closure Rows

No direct NSIT row matched the assigned legacy papers in the first-pass matrix; validator matching by object name remains a receipt-attachment requirement.

Source Integration Summary

The legacy source and internal reasoning supplement are treated as source evidence rather than manuscript prose. Their contributions enter this paper as claim-lane entries, receipt or validator references, finite-domain statements, and limitation boundaries. Speculative or cross-domain interpretations remain separate from closed local results until an explicit adapter, citation, normal-form derivation, calibration row, or falsifier is attached.

Claim Binding Queue

This queue records the evidence discipline for claim strengthening. It separates claims that already have support from residues that remain in falsifier or open-obligation lanes.

Queue	Promote now	Bind next	Residue
none directly assigned	Source routing and claim lanes identify paper-local binding actions.	Verifier catalogs and CAM/crystal routes provide object-name evidence candidates.	Unbound claims remain in falsifier/open-obligation lanes.

Binding Discipline

Each queue item is represented as a delta:

```
local claim at paper time
-> attached later source/tool/receipt/theorem
-> revised representation
-> invariant boundary and remaining residue
```

This preserves the sequential story while allowing later tools, crystals, and standard formalisms to return through the proper lane.

Claim Delta Ledger

This ledger applies the paper's claim-binding queues as explicit back-application deltas. It keeps the sequential paper story intact while allowing later receipts, standard formalisms, CAM/crystal routes, and validators to sharpen the claim.

Delta	Local claim at paper time	Later evidence	Revised representation	Boundary kept
Search By Object Name	No direct reasoned-closure queue was assigned from the first queue map.	Search source routing, verifier catalog, CAM/crystal routes, and paper-local object names.	Create a specific binding queue when an object-name match finds a validator, receipt, theorem, or calibration route.	Until then, claims remain in their existing lanes.

The revised representation records the strongest claim supported by the current evidence package. Promotion into theorem or proposition language occurs only when the named evidence is attached in the manifest or workbook replay.

Source-Backed Evidence Summary

Routed archive and mirror cards are treated as provenance summaries rather than body text. They identify candidate receipts, validators, theorem registries, or supplemental source routes. A card strengthens this paper only when its contribution is bound to a claim lane, evidence object, and boundary.

Evidence card	Score	Publication contribution	Boundary
CQE-paper-04.25: Paper 4.25 - Toolkit for Boundary Repair	81	Paper 4.25 describes the review tools for boundary repair. The tools help a reader construct and inspect repair rows. They do not add claims beyond the Paper 4 theorem. The toolkit works with: ``text correction state...	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.
FINAL_FORMAL_PAPER: Complete Formal Claims: The Folded Strand	75	We present the complete closed-form claim set of the CQE_CMPLX corpus: - 33 papers = 33 folding operations on a self-complementary strand - 144 tools = cumulative analog kit = digital twin surface - 135 digital twins ...	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.
SUMMARY-V-32-Theorems-Registry: Summary Paper V — The 32 Theorems in One Table: Closed-Form Registry	75	This paper is the complete closed-form registry of all 32 theorems in the CQE_CMPLX corpus. Each theorem is stated precisely, given its formal context (where it is proven), and listed with its verifier. The table ...	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.
CQE-paper-02.50: Paper 2.50 - Correction Surface Claim Contract	73	Paper 2.50 lets later papers import the correction surface honestly. It keeps the finite theorem available while preserving the distinction between residue, obligation, and proof.	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.
FORMAL: Paper 01 — Side-Flip SU(2) Doublet (T_BIJECTIVE)	73	PROVEN by structural identification on the T3 chart/J■(O) isomorphism.	Source-backed support; promotion requires the paper-local claim lane and evidence boundary.
Additional evidence cards	23 total	The complete card inventory is maintained in the archive evidence matrix.	Cards are source-discovery aids until bound to paper-local evidence.

Journal Apparatus

Article Type And Intended Venue Fit

This manuscript is written as a formal-methods and mathematical-systems paper inside the series *Formalizing LCR, Unifying Standard Models*. Its intended reader is a technical reviewer who needs claim boundaries, reproducibility routes, and cross-paper dependencies to be visible without relying on private context.

Structured Contribution Statement

Item	Statement
Publication ID	FLCR-05
Legacy source slot(s)	04
Ribbon role	order-1 slot-4: type the boundary, repair row, or curvature-demand condition
Proof form	typed boundary row and next-state admissibility
Received state	state emitted by prior slot 03 and reopened at original slot 04 (Boundary Repair)
Produced state	formal result of original slot 04 plus the same-family lift path toward slot 14

Claim-Evidence Table

Claim	Lane	Evidence to attach	Boundary
Theorem 5.1	receipt_bound_internal_result	Receipt, source card, validator, citation, or CAM route named in the paper manifest	A valid boundary repair converts a failed join into a typed proof-obligation row while preserving the relevant coordinates.

Claim	Lane	Evidence to attach	Boundary
Proposition 5.2	receipt_bound_internal_result	Receipt, source card, validator, citation, or CAM route named in the paper manifest	Repair is idempotent on already-normalized rows.
Protocol 5.3	normal_form_result	Receipt, source card, validator, citation, or CAM route named in the paper manifest	Untyped failure cannot be consumed by later papers as repair evidence.

Figure Plan

Figure	Purpose	Caption
FLCR-05-F1	Typed boundary/admission diagram	Schematic showing how FLCR-05 turns its received state into the produced state while preserving claim lanes and residue boundaries.
FLCR-05-F2	Evidence routing map	Diagram of source papers, archive cards, CAM/crystal routes, validators, and workbook replay paths used by this manuscript.
FLCR-05-F3	Same-family lift context	Diagram placing this paper in the original 00 - 79 ribbon family and showing predecessor/successor slots.

In-Text Figure FLCR-05-F1: Typed boundary/admission diagram

```

received state
-> Typed boundary/admission diagram
-> formal claim lane
-> evidence object / receipt / citation
-> produced state
-> remaining residue

```

Table Plan

Table	Purpose
FLCR-05-T1	Boundary verdict table
FLCR-05-T2	Claim-lane/evidence-boundary table
FLCR-05-T3	Archive evidence card and source-backed upgrade table
FLCR-05-T4	Workbook replay and falsifier table

Worked Example

Classify a candidate as admitted, repaired, boundary, rejected, or calibration-required. The example must name the input object, the operation, the evidence object, the allowed revised claim, and the remaining boundary.

Nomenclature And Glossary

Term	Paper-local meaning
CAM	Defined by this paper as part of the boundary_action proof role; final definition is recorded in the paper-local glossary.
admission	Defined by this paper as part of the boundary_action proof role; final definition is recorded in the paper-local glossary.
boundary	Defined by this paper as part of the boundary_action proof role; final definition is recorded in the paper-local glossary.
claim lane	Defined by this paper as part of the boundary_action proof role; final definition is recorded in the paper-local glossary.
crystal	Defined by this paper as part of the boundary_action proof role; final definition is recorded in the paper-local glossary.
falsifier	Defined by this paper as part of the boundary_action proof role; final definition is recorded in the paper-local glossary.
receipt	Defined by this paper as part of the boundary_action proof role; final definition is recorded in the paper-local glossary.
repair row	Defined by this paper as part of the boundary_action proof role; final definition is recorded in the paper-local glossary.

Appendix FLCR-05-A: Evidence Package

The appendix records:

- 1. Legacy source excerpts or source-card references used in the final claim ledger.
- 2. Receipt and verifier paths, including pass/fail/partial status.
- 3. CAM/crystal query or projection rows used by the paper, with source identity and boundary.
- 4. Standards-window and NSIT evidence used for conformance or routing.
- 5. Falsifier, calibration, or external-data rows that remain unresolved.

Data And Code Availability

The publication package contains the canonical Markdown sources, manifests, source-routing matrices, validation reports, and generated PDF/TeX outputs.

Code and verifier references are cited through manifests, archive evidence cards, receipt catalogs, or workbook replay tables. External datasets or standards citations must be attached before any calibration-bound claim is promoted.

Reviewer Checklist

Check	Reviewer question
Claim lane	Does every strong claim declare its lane and evidence type?
Boundary	Does the paper preserve what it does not prove?
Reproducibility	Is there a receipt, verifier, source card, citation, or replay table for the claim?
Cross-paper import	Does the paper cite the prior FLCR/OPR slot before consuming its result?
Interpretation	Does the analog layer preserve the addressed state while changing labels/access paths?

Declarations

No human-subjects, clinical, or animal-research protocol is asserted by this paper. External empirical claims require separate dataset, calibration, or domain-validation attachments. Competing-interest and funding statements are recorded in the submission metadata.

11. Publication Integration Addendum

11.1 Role In The Series

FLCR-05 (Typed Boundary Repair) occupies the `boundary_action` role at lift depth 0. The paper receives state emitted by prior slot 03 and reopened at original slot 04 (Boundary Repair). It produces formal result of original slot 04 plus the same-family lift path toward slot 14. The state transition is: type the boundary condition that decides whether the state repairs, reflects, curves, or routes onward.

The paper-local proof form is:

`typed boundary row and next-state admissibility`

The integration result is a typed boundary/admission table for legal next-state repair. This addendum records how that result is consumed by the publication series without relying on editorial instructions or private working context.

11.2 Formal Carrier

Formal carrier	Function
boundary value problems	Formal carrier for the paper-local result.
typed interfaces and admissibility conditions	Formal carrier for the paper-local result.
topological boundary/interior distinction	Formal carrier for the paper-local result.
falsifier rows for failed promotions	Formal carrier for the paper-local result.

The LCR, CAM, crystal, forge, and analog vocabulary functions as a typed access layer over these carriers. A relabeling is admissible only when the addressed object, evidence lane, boundary, and downstream import rule remain attached.

11.3 Claim Ledger

Claim	Lane	Statement
Theorem 5.1	receipt_bound_internal_result	A valid boundary repair converts a failed join into a typed proof-obligation row while preserving the relevant coordinates.
Proposition 5.2	receipt_bound_internal_result	Repair is idempotent on already-normalized rows.
Protocol 5.3	normal_form_result	Untyped failure cannot be consumed by later papers as repair evidence.

These claims are consumed at their stated lane. A stronger downstream statement creates a new claim envelope with its own evidence object.

11.4 Evidence Package

Evidence class	Routed items	Status
Legacy sources	04_Boundary_Repair.md, supplements/04_INTERNAL_REASONING_SUPPLEMENT.md	routed evidence
Evidence inputs	CAM_CLAIM_100_SCORE_AUDIT.md, NSIT_TOOL_CLOSURE_MATRIX.md, NSIT_REASONED_CLOSURE_PASS.md, PAPER_REWORKS_CRYSTAL_PROJECTION.json	routed evidence
CAM/crystal sources	PAPER_REWORKS_CRYSTAL_PROJECTION.json, AGENT_CRYSTAL_WORK_HARVEST.json, runtime standards artifact, notebook-derived source bundle	routed evidence
Standards-window inputs	window_summary.json, node DBs, window DBs	routed evidence
Receipts	external requirement	not attached in this source package
Validators	external requirement	not attached in this source package

Standards-window, runtime, and notebook-derived artifacts are treated as evidence-routing surfaces. They support source discovery, conformance checking, and reapplication, but they do not replace citations, receipts, normal-form derivations, calibration rows, or falsifiers.

11.5 Results Representation

The paper's results section is organized around five publication objects:

1. the exact object acted on at this slot;
2. the admissible operation or transformation;
3. the receipt, enumeration, derivation, lookup, or external theorem supporting the operation;
4. the residue or boundary left after the result;
5. the downstream import rule.

11.6 Figures And Tables

Figure	Role
FLCR-05-F1: Typed boundary/admission diagram	Visualizes the proof object, routing, or boundary.
FLCR-05-F2: Evidence routing map	Visualizes the proof object, routing, or boundary.
FLCR-05-F3: Same-family lift context	Visualizes the proof object, routing, or boundary.

Table	Role
FLCR-05-T1: Boundary verdict table	Records claim lanes, evidence, sources, or falsifiers.
FLCR-05-T2: Claim-lane/evidence-boundary table	Records claim lanes, evidence, sources, or falsifiers.

Table	Role
FLCR-05-T3: Archive evidence card and source-backed upgrade table	Records claim lanes, evidence, sources, or falsifiers.
FLCR-05-T4: Workbook replay and falsifier table	Records claim lanes, evidence, sources, or falsifiers.

11.7 Limits And Falsifiers

The paper proves only the object and domain declared in its claim ledger. Physical, Standard Model, observer, calibration, or synthesis claims require the additional lane evidence stated by their destination papers. CAM/crystal and forge language is operational only when represented as an address, lookup, receipt, validator, or reapplication path.